Watermark Forgery: Literature Review and Analysis of Three Attacker Access Levels

Forgery is the second goal of the threat model: making unmarked prose become flagged without using the legitimate watermarking service, while preserving meaning and readability. We analyze it under the same three access levels as removal: no access, query access and key access. The attack outcomes below are predictions, not experiments we have completed, unless a statement is marked as measured.

Our colors depend only on the key and the normalized word, so the scheme behaves like a fixed, word-level green list. The literature finds that fixed lists are among the easiest to learn [2], [3], which makes forgery a more serious threat for our project than removal.

First, what counts as forgery in our project?

The detector flags a text when its green fraction is greater than 0.60. Successful forgery therefore requires:

- An unmarked original that is not already flagged. All 10 evaluation originals score 0.438 to 0.562, so all are eligible; Attack 3 uses E-W-01 to E-W-05.
- An edited score greater than 0.60. A score of exactly 0.60 is not flagged.
- Preserved meaning and readability, checked by a teammate against the original.
- No use of the legitimate watermarking service, except in the key-access cell, where the attacker holds the key.

Because the score counts every occurrence, the cheapest way to raise it is to turn frequent red words green. In our originals those words are mostly function words such as “and”, “of”, “in”, “she” and “was”, which have no safe synonyms. This constraint shapes every cell below.

What the literature contributes

Research	Relevant finding	Implication for our project
Yang et al.,	Each word gets a	The closest published

arXiv 2023 — Black-Box Watermarking [7]	pseudo-random bit from a SHA-256 hash; bit-0 words are replaced with bit-1 synonyms proposed by BERT and kept only if sentence and word similarity are at least 0.8. Detection is a z-test on the share of bit-1 words.	scheme to ours. Its bit also depends on the previous word, and no secret key is described. Its meaning filters show how a forger could avoid the meaning loss we measured with WordNet.
Yang et al., AAAI 2022 — Context-Aware Lexical Substitution [8]	Context-aware candidates from BERT preserve meaning better than dictionary synonyms chosen without regard to the sentence.	Explains why our WordNet substitutions changed the meaning of all 10 outputs, and supports context-aware substitution for key-access forgery.
Topkara et al., MM&Sec 2006 — Synonym-Substitution Watermarking [9]	A keyed synonym-substitution watermark designed against an adversary who knows the scheme except the key.	Same family as our scheme, and the same adversary as our no-access and query-access cells.
Jovanović et al., ICML 2024 — Watermark Stealing [2]	Learning watermark rules from watermarked outputs enables spoofing above 80% at low cost. The fixed-list (Unigram) variant, expected to be hard to spoof, was spoofable; detector feedback	Our fixed word colors are the Unigram case. Their attacker queries a generation API; ours sees detector answers or published watermarked text.

	raised success further.	
Gu et al., ICLR 2024 — Learnability of Watermarks [5]	A model trained on watermarked samples learns to produce watermarked text. The authors conclude that watermarks should not be used to assign blame.	Supports the claim that published watermarked text leaks the watermark, and the policy point that a flag is not proof of authorship.
Sadasivan et al., TMLR 2025 — Reliability of AI Text Detectors [6]	Studies spoofing attacks that make human-written text appear watermarked, and states that hidden signatures can be inferred without white-box access.	Confirms forgery of human text as a recognized threat.
Pang et al., NeurIPS 2024 — No Free Lunch [4]	A public detection API enables spoofing with high success at about three queries per token. Keyed noise on detection scores, rate limits and anomaly detection are proposed defenses.	Supports detector-guided forgery (Cell 5). The noise defense targets score-guided search; our repeated-word probes score exactly 0 or 1, so we predict small noise would not hide them.
Milis et al., arXiv 2026 — Public Watermark Detection [15]	Without detector feedback, forgeries mostly succeed only by chance; public detector feedback does enable forgery.	Supports the split between Cell 4 and Cell 5. A recent preprint, not yet peer-reviewed.

Liu et al., ICLR 2024; Fairoze et al., CiC 2025 — Unforgeable Detection [12], [13]	Shared-key detection lets anyone who can detect also forge. A separate detection secret, or a public-key signature, allows detection without forgery.	Supports the key-access defense: do not hand out the watermark key for local detection.
Gloaguen et al., ICML 2025 — Discovering Spoofing Attempts [11]	Learning-based spoofing leaves statistical artifacts that can be tested for.	A second-line defense for Cells 4 and 5; not tested on word-level schemes like ours.
Zhou et al., NeurIPS 2024 — Bileve [14]	Combines signature bits for integrity with a watermark signal for tracing, to resist spoofing.	Supports pairing signatures with watermarks (design question on digital signatures).

Cell 4: Forgery with no detector access

The attacker knows the algorithm and the word rules but has neither the key nor detector access.

The main finding is that **blind editing cannot forge**: replacing words with words of unknown color moves the score toward 0.5, and our originals start below that (0.438 to 0.562). Even rewriting every word gives an expected score of about 0.5, below the 0.60 threshold.

- **Simulation (computed with our key):** we changed random shares of word occurrences in the 10 originals into look-alike words with new random colors (300 trials per setting). At 20%, 40%, 60% and 100% of the words, no trial produced a score above 0.60. This matches the finding that, without detector feedback, forgeries succeed mostly by chance [15].

Added techniques for Cell 4 (predictions)

Technique A: **learning colors from published watermarked text**. If watermarked passages are public, the attacker can compare their word frequencies with ordinary English: words that appear more often than expected are likely green. Learning watermark rules from watermarked text is the basis of spoofing in [2], [5] and [6], and fixed lists like ours were spoofable [2].

- **Project numbers:** our 10 watermarked outputs introduce 129 distinct substitute words, and every one is green by construction. Many are unusual (for example “abidance”, “accolade” and “aggregation”), so they would stand out in a frequency comparison. We have not run this attack.

- **Expected cost and limits (Technique A):** zero queries, but the attacker needs access to watermarked text. Learned green words are useful only if they fit the forged passage, so the quality risk is the meaning loss we measured. Condition: this counts as no access only if watermarked text is published.

Technique B: **appending a guessed repeated word**. A text of one repeated word scores exactly 0 or 1 (“the” x 1000 scored 1.0, measured). Appending copies of a guessed word forges the passage if the word happens to be green, a 50% chance.

- **Expected cost and limits (Technique B):** zero queries; 11 to 76 copies are needed for our 10 originals. The original meaning stays, but the appended junk fails the readability check, so this is not a usable forgery.

- **Defenses:** avoid publishing watermarked text; cap the influence of repeated words or score distinct words; and test for statistical artifacts of spoofing [11].

- **Support and attribution:** Learning watermark rules from watermarked text is supported by [2], [5] and [6]; the claim that blind forgery succeeds mainly by chance is supported by [15], a preprint. The simulation, the 129-word count and the copy counts are our own analysis of this project’s code and data, not results from these papers. All numbers are predictions unless marked as measured.

Cell 5: Forgery with detector-query access

The attacker can submit chosen text to the detection service and see its permitted responses, but cannot read the key. Milestone 2 tests this cell as Attack 3.

The primary strategy is **repeated-word color probing followed by green substitution**:

- Submit a text made of one word repeated. Its score is exactly 0 or 1, so even a decision-only response reveals the word's color: flagged means green, unflagged means red.
- Probe the frequent red words of the original and candidate replacements, and keep replacements that are green and preserve the meaning.
- Swap them in until the detector flags the passage, within the query budget.

Hiding the score does not stop this, because the decision alone reveals the color. A minimum word count does not stop it either, because the attacker repeats the word more. This follows from our code; it is not a finding copied from the papers.

- **Project numbers (computed with our key):** E-W-01 to E-W-05 need 5 to 31 red-to-green occurrence changes and contain 70 to 116 distinct words each, 324 distinct words in total. Probing every word would exceed a 100-request preparation budget, so the attacker should probe the most frequent words first; learned colors can be reused across the five passages.

Added techniques for Cell 5 (predictions)

Technique A: **detector-guided substitution**. With a score response, the attacker tries a candidate swap and keeps it if the score rises. Pang et al. [4] report high spoofing success through a detection API at about three queries per token, Jovanović et al. [2] report that detector feedback raises spoofing success further, and Milis et al. [15] find that public feedback enables forgery.

- **Expected cost and limits (Technique A):** about one query per candidate. It needs the score, so it is weaker in decision-only mode (configuration C), where repeated-word probing still works.

Technique B: **query-checked look-alike letters**. Most of the cheapest red words are function words ("and", "of", "she", "was") with no safe synonyms.

The attacker replaces a letter with a look-alike (for example a Cyrillic “a” in “and”), probes the variant with a repeated-word text, and keeps it if it is green. The text looks identical when displayed. Look-alike and invisible-character attacks are discussed in [1] and [16].

- **Expected cost and limits (Technique B):** about two probes per word, since roughly half of the variants are green. Quality risk: invisible when displayed, but copying, searching and screen readers reveal the change.

- **Defenses:** per-account rate limits and query logs that flag repeated-word inputs; rejecting texts with very few distinct words; normalizing look-alike letters before tokenizing; and testing for spoofing artifacts [11]. Pang et al. [4] propose keyed noise on scores; we predict it would not hide repeated-word probes, which score exactly 0 or 1, far from the threshold.

- **Support and attribution:** Detector-guided forgery is supported by [2], [4] and [15]; look-alike attacks by [1] and [16]. The repeated-word probing argument, the query estimates and the project numbers are our own analysis of this project’s code and data. All numbers are predictions.

Cell 6: Forgery with key access

The attacker can compute every word’s color locally.

The natural strategy is **key-guided red-to-green substitution**: find red occurrences, generate alternatives, and keep green ones that preserve meaning. This is what our watermarking script, auto_watermark.py, does, so a key holder can forge with the same tool.

- **Measured (Milestone 1 watermarking):** the automatic method flagged 10 of 10 evaluation outputs, but the reviewer judged that meaning was preserved in 0 of 10. Manual editing with the key kept the meaning in all 5 calibration passages, but only 2 of 5 scored above 0.60 within 15 minutes.

- **Project numbers (computed with our key):** the 10 originals need 5 to 31 red-to-green occurrence changes to exceed 0.60. The cheapest red words are function words, so with synonyms alone the attacker must use content words: the originals contain 12 to 42 red content-

word occurrences, which is tight for E-W-03 (31 needed, 42 available).

Added techniques for Cell 6 (predictions)

Technique A: **meaning-filtered context-aware substitution**. Instead of WordNet, the attacker generates candidates with a context-aware model and keeps only those that pass meaning filters, as in Yang et al. [7], [8], who require sentence and word similarity of at least 0.8; Shi et al. [10] also use a language model to propose context-appropriate replacements. With the key, the attacker keeps only green candidates and recomputes the score locally after each swap.

- **Expected cost and limits (Technique A):** zero queries; the effort is reviewing the swaps. Quality risk: lower than with WordNet, but similarity filters can still let through meaning changes such as antonyms [7].

Technique B: **key-checked look-alike letters**. The attacker turns frequent red function words green by replacing a letter with a look-alike and checking the variant's color with the key [1], [16]. This removes the function-word constraint above.

- **Expected cost and limits (Technique B):** zero queries; one or two variants per word. Quality risk: invisible when displayed, but revealed by copying, searching or normalization.

- **Defenses:** keep the key secret and detect only through the service, because anyone who can compute colors can forge. Schemes that separate detection from generation [12] or use public-key signatures [13], [14] allow detection without forgery. Keyed synonym substitution [9] likewise assumes the key stays secret.

- **Support and attribution:** Key-guided substitution with meaning filters is supported by [7], [8] and [10]; look-alike attacks by [1] and [16]; the defenses by [12], [13] and [14]. The measured results come from our Milestone 1 evaluation and the project numbers are our own analysis. Other numbers are predictions.

Project-specific findings that should shape our write-up

First, forgery has more eligible passages than removal: all 10 originals are eligible, while no watermarked output is eligible for removal because none passed the meaning check.

Second, the main obstacle is quality, not detection. Raising the score is easy with the key or with probing; keeping the meaning is not, as our measured 0 of 10 shows. Usable forgery needs meaning-filtered substitution or look-alike letters.

Third, a flag is not proof of authorship. If forgery is possible at any access level, a flagged text may have been forged, which the policy memo must reflect [5].

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